

Course Write-Up (for Courses other than External Degree Programmes)

Key items	PEIs' Inputs												
Course Title: 1. If it is 100% e-learning course, add "(e-learning)" at the end of the course title. 2. If conducted in Mandarin, add "(Mandarin)" at the end of the title. 3. Include prefix "(MC)" for stackable certificate courses.	Advanced Certificate in Enterprise Blockchain Solution Architecture												
Course Developer	☒ Self-developed ☐ Externally-developed ☐ Jointly-developed <i>(Curriculum self-developed by Tertiary Infotech.)</i>												
Qualification to be Awarded upon Course Completion <i>Note: The name of qualification to be awarded should be the same as the course title.</i>	Advanced Certificate in Enterprise Blockchain Solution Architecture												
Qualification to be Awarded by:	Tertiary Infotech Academy												
Brief Description of Course (including learning objectives for each module)	Brief description of course and learning outcomes: The Advanced Certificate in Enterprise Blockchain Solution Architecture equips learners with practical skills to analyse business requirements, design decentralised and permissioned ledger solutions, operate blockchain network components, and evaluate platform choices for enterprise use cases. Upon successful completion, learners will demonstrate the ability to explain distributed ledger concepts; design secure, scalable blockchain architectures; select suitable development tools, node infrastructure, consensus models and storage patterns; compare major enterprise blockchain platforms; apply governance and risk controls; and produce an implementation-ready blockchain solution blueprint. The course comprises five modules. Each module consists of 10 sessions: 9 teaching sessions and 1 assessment session. The modules are delivered through live online classes, guided demonstrations, hands-on labs and applied design activities. The learning objective for each module is as follows: <table border="1"> <thead> <tr> <th>S/N</th><th>Module Title</th><th>Learning objective</th></tr> </thead> <tbody> <tr> <td>1</td><td>Module 1: Distributed Ledger and Blockchain Core Concepts</td><td>Learners will be able to explain the purpose of distributed ledgers in enterprise environments; describe blocks, transactions, hashing, consensus, wallets, tokens and smart contracts; compare public, private and consortium network models; and identify common enterprise blockchain use cases across finance, supply chain, identity, records management and digital assets.</td></tr> <tr> <td>2</td><td>Module 2: Blockchain Architecture and Enterprise Solution Design</td><td>Learners will be able to describe the responsibilities of a blockchain solution architect; gather stakeholder, business, technical and governance requirements; design network topology, identity, access control, consensus, data flow and integration patterns; evaluate scalability and performance considerations; and produce an implementation-ready enterprise blockchain solution blueprint.</td></tr> <tr> <td>3</td><td>Module 3: Blockchain Development Infrastructure and Node Operations</td><td>Learners will be able to identify suitable blockchain tools, frameworks, wallets, explorers, APIs, SDKs and deployment utilities; set up and configure blockchain network nodes; explain</td></tr> </tbody> </table>	S/N	Module Title	Learning objective	1	Module 1: Distributed Ledger and Blockchain Core Concepts	Learners will be able to explain the purpose of distributed ledgers in enterprise environments; describe blocks, transactions, hashing, consensus, wallets, tokens and smart contracts; compare public, private and consortium network models; and identify common enterprise blockchain use cases across finance, supply chain, identity, records management and digital assets.	2	Module 2: Blockchain Architecture and Enterprise Solution Design	Learners will be able to describe the responsibilities of a blockchain solution architect; gather stakeholder, business, technical and governance requirements; design network topology, identity, access control, consensus, data flow and integration patterns; evaluate scalability and performance considerations; and produce an implementation-ready enterprise blockchain solution blueprint.	3	Module 3: Blockchain Development Infrastructure and Node Operations	Learners will be able to identify suitable blockchain tools, frameworks, wallets, explorers, APIs, SDKs and deployment utilities; set up and configure blockchain network nodes; explain
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1	Module 1: Distributed Ledger and Blockchain Core Concepts	Learners will be able to explain the purpose of distributed ledgers in enterprise environments; describe blocks, transactions, hashing, consensus, wallets, tokens and smart contracts; compare public, private and consortium network models; and identify common enterprise blockchain use cases across finance, supply chain, identity, records management and digital assets.											
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3	Module 3: Blockchain Development Infrastructure and Node Operations	Learners will be able to identify suitable blockchain tools, frameworks, wallets, explorers, APIs, SDKs and deployment utilities; set up and configure blockchain network nodes; explain											

			node synchronisation, connectivity, monitoring and resilience requirements; and apply operational practices for maintaining distributed ledger infrastructure.
	4	Module 4: Governance, Security, Risk and Decentralised Data Controls	Learners will be able to identify technical, operational, regulatory and project risks in blockchain deployments; define security and governance controls; compare off-chain, decentralised and hybrid storage approaches; explain data integrity, privacy, retrieval and cost considerations; and recommend mitigation measures for responsible enterprise implementation.
	5	Module 5: Enterprise Blockchain Platform Architectures	Learners will be able to evaluate Ethereum-based architectures, Hyperledger Fabric permissioned ledgers, Corda workflow networks, Cardano ecosystem architecture and Quorum private Ethereum networks; compare platform-specific identity, permissioning, privacy, consensus, smart contract and workflow models; assess design trade-offs; and recommend suitable platform and integration approaches for enterprise use cases.
Target Audience	Primary Target Audience IT professionals, software engineers and solution architects who want to design blockchain-enabled enterprise systems. Developers seeking to move from application development into blockchain architecture, decentralised applications or smart contract solution design. System, cloud, DevOps and network engineers who need to understand node infrastructure, deployment models and operational requirements for distributed ledger networks. Business analysts, product managers and digital transformation professionals responsible for evaluating blockchain use cases and translating requirements into technical solutions. Risk, compliance, audit and governance professionals who need a practical understanding of blockchain architecture, platform risks and control considerations. Entrepreneurs, consultants and project leads planning blockchain pilots, proof-of-concepts or enterprise implementations. Fresh graduates from IT, computer science, data science, engineering or related programmes who want career-ready blockchain solution architecture skills.		
(For certificate and above qualifications only) – Potential Employment Opportunities upon Course Completion	Graduates of the Advanced Certificate in Enterprise Blockchain Solution Architecture can pursue a range of roles in both the public and private sectors, including: Blockchain Solution Architect Enterprise Blockchain Architect Blockchain Developer / Engineer Smart Contract Solution Designer Distributed Ledger Technology Consultant Web3 Solution Consultant Blockchain Infrastructure Engineer Blockchain Project Analyst Digital Asset Technology Analyst Enterprise Technology Consultant Innovation / Digital Transformation Analyst Technical Product Manager for Blockchain Solutions These roles span industries such as finance, fintech, supply chain, healthcare, government, technology, digital identity and enterprise software, with opportunities for progression into senior architecture, consulting and technology leadership positions.		
Does the proposed course curriculum follow a foreign or international curriculum through full-time programme at the Primary, Secondary, and/or High School levels (grades 1-12), including self-developed curriculum?	☐ Yes ☒ No <i>If yes, please state the location of PEI's premise for the course to be conducted in: N/A</i>		

Course Details	Mode of Delivery: ☐ Face-to-face ☐ Blended ☒ E-learning (100% synchronous live virtual classes) Duration: ~17 weeks / ~4 months (part-time) Class Frequency: 3 days per week x 3 hours per day (part-time), 7:00 PM — 10:00 PM Total Course Hours: 150 hours (part-time) Computation for course hours: 5 modules x 10 sessions x 3 hours = 150 hours. Each module consists of 10 sessions: 9 teaching sessions and 1 assessment session.						
Minimum Entry Requirements	Age: At least 21 years old Academic: At least C6 at GCE O-Level in any 3 subjects, or equivalent; or Mature candidate who is at least 25 years old with at least 4 years of working experience						
(For certificate qualifications only) – to be read in conjunction with SSG Advisory 3/2022	Will the PEI issue modular certificate(s) that students can stack towards the award of the Certificate Qualification? ☒ YES ☐ NO <i>Note: Only "Terminal" Certificate Courses can be registered as Modular Courses that stack to a Certificate Qualification.</i> <table border="1" style="width: 100%;"> <thead> <tr> <th>S/N</th><th>Title of certificate qualification</th><th>Modules</th></tr> </thead> <tbody> <tr> <td>1.</td><td>Advanced Certificate in Enterprise Blockchain Solution Architecture</td><td> 1. Module 1: Distributed Ledger and Blockchain Core Concepts 2. Module 2: Blockchain Architecture and Enterprise Solution Design 3. Module 3: Blockchain Development Infrastructure and Node Operations 4. Module 4: Governance, Security, Risk and Decentralised Data Controls 5. Module 5: Enterprise Blockchain Platform Architectures </td></tr> </tbody> </table> Maximum time allowed to complete the modular courses for award of the certificate qualification: ~4 months (part-time). Terms and conditions to be met to be awarded the Certificate: To be awarded the certificate, participants must: • Achieve a pass in all required assessments (one per consolidated module), and • Maintain a minimum attendance of 75% throughout the course.	S/N	Title of certificate qualification	Modules	1.	Advanced Certificate in Enterprise Blockchain Solution Architecture	1. Module 1: Distributed Ledger and Blockchain Core Concepts 2. Module 2: Blockchain Architecture and Enterprise Solution Design 3. Module 3: Blockchain Development Infrastructure and Node Operations 4. Module 4: Governance, Security, Risk and Decentralised Data Controls 5. Module 5: Enterprise Blockchain Platform Architectures
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Articulation Pathway	Is the proposed course a pathway programme? ☐ YES ☒ NO						
Information on External Course Developer (for externally developed courses only)	Not applicable — the course is self-developed by Tertiary Infotech Academy.						
Course Accreditation	Is the proposed course accredited by any external organisation? ☐ YES ☒ NO						
Association, Collaboration or Affiliation	Is there any other form of association, collaboration or affiliation with any other organisation or persons, either local or foreign, in respect of this course? ☐ YES ☒ NO						
Courses with Industrial Attachment	Does the proposed course have industrial attachment? ☐ YES ☒ NO						
Other Information, if applicable	N.A.						

I, **Dr. ANG CHEW HOE**, (NRIC/Passport no. S1808997A) manager of **TERTIARY INFOTECH ACADEMY PTE. LTD.** declare that:

- (a) the course has been approved by the Academic Board;
- (b) the Academic Board has approved the deployment of teachers for the course in accordance with Regulation 26 of Private Education Regulations;
- (c) the assessment methods and procedures for the course have been approved by the Examination Board; and
- (d) all information provided in this application is true and accurate to my knowledge.

Alfred Ang

19 Jun 2026

Date: _____

(Signature of manager & date)